



PAH formation from jet stirred reactor pyrolysis of gasoline surrogates

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ABSTRACT

Soot particles and their precursor polycyclic aromatic hydrocarbon (PAH) species, formed during combustion, are responsible for particulate emissions in gasoline direct injection (GDI) engines. To better understand the effects of fuel composition on formation of soot in GDI engines, the pyrolysis of several gasoline surrogates was studied in a jet-stirred reactor across a broad temperature range at atmospheric pressure and 1 s residence time. Fuel and intermediate species, including aromatics up to naphthalene, were measured using gas chromatography (GC). PAH concentrations from pyrolysis of surrogate fuels were compared to gain insight into the effects of fuel composition on PAH formation. In addition, synergistic effects were observed in pyrolysis experiments of binary blends. A detailed kinetic model, recently developed at Lawrence Livermore National Laboratory (LLNL), successfully captured the effects of blending and the concentration of major PAHs. Major reaction pathways are discussed, as well as the role of important intermediate species, such as acetylene, and resonantly stabilized radicals such as allyl, propargyl, cyclopentadienyl, and benzyl in the formation of PAH.

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1. Introduction

Soot particles and their precursors (polycyclic aromatic hydrocarbons (PAHs)), emitted from on-road vehicles, contribute to anthropogenic aerosol emissions, harming both the environment and human health [1,2]. In recent years, stringent regulations against NO_x and soot emissions have increased the need for better models to predict pollutant formation from engines. Emission of these carcinogenic pollutants is expected to increase with the growing demand for light duty vehicles powered by gasoline direct injection (GDI) engines. Because of its fuel efficiency, this type of engine is widely used in the transport sector; however, they are characterized by a higher tendency to form soot. GDI engines are reported to produce more soot particles than conventional port fuel injected spark injection (PFI-SI) engines, mainly because of poor in-cylinder fuel/air mixing, which leads to fuel rich regions in the combustion chamber and localized high combustion temperatures [3,4]. Meanwhile, GDI engines produce more ultrafine and nano-sized particles than compression ignition (CI) engines [5], and smaller particles are more harmful to the environment and human health.

Clarifying the formation mechanisms of soot and their PAHs precursors in combustion has been the target for decades of re-

search [6–9]. Fundamental data from well-defined reactors, and kinetic modeling with detailed reaction pathways, are promising tools to bridge the gap [10,11]. The combustion properties of molecules representative of those in gasoline, such as normal-alkanes, branched alkanes, and aromatics, have been extensively investigated, providing data to develop detailed reaction mechanisms for PAHs. However, the experimental study of soot and PAH formation by gasoline surrogates is limited, and most of the experiments were performed with flames [12–15] or shock tubes [16–19]. Park et al. [15] studied the PAH formation of gasoline surrogates by PLIF, but only qualitative PAH data were obtained. Li et al. [12], Inal et al. [13] and Bakali et al. [14] quantitatively investigated the PAH in laminar premixed flames diagnosed by TOF MS and GC-MS. However, the uncertainties of the measurement close to the burner were substantial because of interactions between the sampling probe and the burner surface [13]. In the shock tube, all work has focused on soot yield, and the concentrations of PAH could not be quantified.

A jet stirred reactor (JSR) is considered to be a zero-dimensional reactor, suited for gas-phase kinetic studies. The oxidation of gasoline surrogates in JSR has been studied widely. Chen et al. [20,21] has studied the oxidation of FACE A,C & F gasoline surrogates in JSR. Dagaut et al. [22–25] and Dubreuil et al. [26] did a large amount of research on the oxidation of PRF and TPRF. However, with the exception of a previous study by this group [27], few studies on pyrolysis and PAH kinetics of gasoline surrogates

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have been conducted in a JSR. In our previous work [28], two toluene primary reference fuels (TPRF), comprised of toluene/*n*-heptane/*iso*-octane with RON= 70 and 97.5, were studied during JSR pyrolysis experiments. The results emphasized that propargyl radical and benzyl radical play an important role in PAH formation in pyrolysis of TPRF fuels. To better understand compositional effects on PAH formation, the pyrolysis of additional *n*-heptane/toluene binary mixtures and *n*-heptane/toluene/*iso*-octane ternary mixtures was studied in this work. A comprehensive kinetic model for the gasoline surrogate mixtures was also developed to accurately predict the fuel pyrolysis, along with the formation of PAH.

2. Experimental method

The JSR at KAUST, designed according to the recommendations of Ayass et al. [29], was used for these experiments. The suggestions in [29] recommended greater residence times to ensure a well-mixed environment, therefore, one second residence was chosen for this study. Also, because a JSR with a larger diameter and a crossed nozzle configuration allows for better mixing, a reactor with a crossed nozzle configuration and a volume of either 31.6 cm³ or 76 cm³ was used for the experiments. The nozzle diameter was 0.33 mm. Fuel was vaporized at 500 K and then diluted with nitrogen, also functioning as carried gas, to introduce fuel vapor into the spherical reactor. A K-type thermocouple mounted into the reactor monitored reaction temperature. The products were sampled using a sonic-throat gas sampling probe located at the outlet of the reactor and connected to a mechanical pump. The pressure and temperature drop across the orifice was sufficient to quench reactions in the sample transfer line. The sample gas was analyzed online using two gas chromatographs (GC), an Agilent Refinery Gas Analysis (RGA) and an Agilent 7890B. The Agilent RGA system follows ASTM D1945, D1946, and UOP 539 methods, and uses TCD and FID detectors to quantify H₂ and C₁–C₅ hydrocarbons from TPRF pyrolysis. The Agilent 7890B system is equipped with an Agilent DB-1 column (15 mm × 0.32 mm) for separation and an FID to detect PAHs. The total time of the temperature program was 17.83 mins, with the following oven control: (1) 45 °C for 6 mins, (2) 30 °C /min to 100 °C for 4 mins, (3) 30 °C/min to 280 °C for 0 min. PAHs in the GC spectrum were identified by injecting standard samples (TCL PAH Mix by SUPELCO) into the GC. PAH mole fractions were calculated by calibrating with toluene and applying corresponding effective carbon numbers. The carbon balance was checked for the measurements reported here in. The carbon balance was found to be above 95% for inlet temperatures of 780 – 1050 K where we did not find any evidence of soot formation. But for higher temperatures, formation of soot on reactor walls was noticed, and the carbon recovery from measurements was low. Uncertainty in the reported mole fractions was ±5% for reactants, ±15% for C₁–C₅ pyrolysis products, and ±30% for benzene, alkyl-substituted benzene, and larger PAH's (e.g.: indene, naphthalene). Errors resulted from uncertainty in the mass flow controllers and syringe pump, measurement reproducibility and uncertainties in gas chromatograph measurements. The detailed schematic of the experimental setup can be found in the Supplementary material.

Following our previous study [28], the pyrolysis of TPRF 80 and TPRF90 was performed in a JSR with a volume 76 cm³. Three blends: *Fuel 1*, *Fuel 2*, and *Fuel 3* were also studied, and the relative amounts of *n*-heptane and toluene were varied. In addition, one primary reference fuel (PRF), which had a RON= 70, was also tested in this work. All the *n*-heptane/toluene blends and PRF70 were conducted in the previous generation JSR with a volume of 31.6 cm³. The size difference in the JSRs did not affect the experimental results, as fuel concentration and residence time were

maintained at the same levels in both reactors. JSR conditions were as follows: the fuel mixture and nitrogen concentrations were fixed at 0.25% and 99.75% respectively, pressure at 1 ± 0.1 bar, residence time at 1 ± 0.05 s, and temperatures from 800 K to 1250 K. Only pyrolysis experiments were conducted in this work, so the equivalence ratio is infinite and the test mixtures do not contain oxygen.

3. Kinetic model development

A new PAH mechanism, currently under development at LLNL, was used for simulations in this study. This new mechanism was built hierarchically, based on the recent aromatic mechanism from Kukkadapu et al. [31]; it describes the growth of PAH's by reactions with important intermediates like acetylene, propene, propyne, allene, 1,3 butadiene, vinylacetylene, benzene, naphthalene, methyl, propargyl, allyl, phenyl, naphthyl radicals. The reaction pathways and the associated rate parameters for reaction with these intermediates were adopted from recent *ab initio* studies noted in [25]. Using the different repetitive growth mechanisms, the mechanism describes the formation of PAH's containing up to C₂₄ carbons (e.g.: coronene, ethynyl-naphthotetraphene: structure shown in supplementary material). However, in the present paper, the kinetic model used is limited to the formation of PAH's containing up to four aromatic rings. The PAH mechanism containing up to seven fused rings will be covered in forthcoming publications, in which the mechanism is systematically validated against the speciation profiles of different PAH's produced during pyrolysis of ethylene, acetylene, propene, propyne, allene, butadiene, ptenenes, cyclopentadiene conducted using JSR, diffusion flames, and flow reactors. The validations have been conducted to ensure that the mechanism has successfully predicted the formation of PAHs from C₁–C₅ precursor species and described in [32]. In the present study, the focus is on the chemistry related to benzyl radicals produced from abstractions from the methyl site in substituted aromatics and known to play an important role in PAH growth during pyrolysis and fuel rich oxidation of TPRF fuels.

The important reactions of benzyl radicals toward the growth of large PAHs have been classified into three major categories listed below, and discussed in detail below:

- (a) decomposition of benzyl radicals
- (b) recombination reactions with radical species
- (c) addition reactions with stable intermediates

3.1. PAH growth through decomposition of benzyl radicals

At high temperatures, benzyl radicals can undergo decomposition reactions that produce fulvenallene (C₇H₆) + H or *o*-benzyne + CH₃. Rates for decomposition of benzyl radicals have been adopted from a study by Derudi et al. [33]. *o*-Benzyne is known to be a very reactive intermediate and its role in the growth of PAHs has been studied in earlier work [34–36]. Based on the findings of Comandini and Brezinsky [35], Matsugi and Miyoshi [36], and Friedrichs et al. [34], the mechanism describes the reactions of *o*-benzyne with benzene, naphthalene, acetylene, ethylene, propene, propargyl radical and phenyl radical, resulting in the production of styrene, phenyl acetylene, naphthalene, phenylpropene, indene and fluorene respectively. The mechanism also describes the unimolecular decomposition of benzyne to acetylene and diacetylene (C₄H₂). The rate for this decomposition reaction has been adopted from a study by Lynch et al. [37].

Fulvenallene, on the other hand, is a stable intermediate and its role in the growth of PAHs is via the reactions of resonance-stabilized fulvenallenyl radicals (C₇H₅). As suggested by da Silva and Bozzelli [38], C₇H₅ possesses characteristics similar to propargyl and cyclopentadienyl radicals and it can facilitate formation

of two- and three-ring PAHs through recombination with itself, or by reactions with propargyl (C_3H_3). Based on this suggestion [38], C_7H_5 radicals were modeled to produce naphthalene, biphenyl and phenanthrene by recombination reactions with C_3H_3 , and with itself, respectively. The rates for recombination of C_7H_5 with C_3H_3 and with itself were taken from Miller and Klippenstein's [39] study on self-recombination of propargyl radicals. However, to account for the steric factors, the pre-exponential A factors were lowered by a factor of two for the fulvenallenyl reaction with propargyl, while the self-recombination rate was reduced by a factor of four. Furthermore, the mechanism also describes the reaction of C_7H_5 with acetylene. The reactions pathways and the corresponding rates for this reaction have been adopted from the *ab initio* study of Matsugi and Miyoshi [36].

3.2. PAH growth through recombination reactions with radical species

At low to intermediate temperatures ($600\text{ K} < T < 1200\text{ K}$), where the temperature is not high enough to overcome the activation barrier for unimolecular decomposition, benzyl radicals can be consumed by chain terminating self-recombination reactions, or recombination reactions with methyl, propargyl or allyl radicals. The self-recombination of benzyl radicals results in production of 1,2 diphenyl ethane, commonly referred as bibenzyl ($C_{14}H_{14}$). $C_{14}H_{14}$ undergoes sequential dehydrogenation which results in the production of phenanthrene. This reaction sequence, pertaining to conversion of $C_{14}H_{14}$ to phenanthrene, was investigated recently by Sinha and Raj [40], and it includes the formation of 1,2-diphenylethylene ($C_{14}H_{12}$), commonly referred to as *stilbene*, which—upon further dehydrogenation—results in the production of phenanthrene. The recombination rate of benzyl radicals used in the present study was taken from recent experimental work by Matsugi and Miyoshi [41]. Analogous reactions from the base chemistry were used to describe the conversion of $C_{14}H_{14}$ to stilbene ($C_{14}H_{12}$). The vinylic/aryl $C_{14}H_{11}$ radicals produced after H-abstractions from $C_{14}H_{12}$ experienced a ring closure reaction resulting in phenanthrene. The $C_{14}H_{11}$ radicals from this reaction sequence were also produced during reactions of phenyl (C_6H_5) radicals with phenylacetylene ($C_6H_5C_2H$), which was studied by Aguilera-Iparraguirre and Kloppe [42]; for this reason, the reactions and associated rates from [42] were used to describe the conversion of $C_{14}H_{11}$ to phenanthrene. Because of the similarity between allyl and benzyl radicals, the recombination of the benzyl radical with the allyl radical was also modelled in the present mechanism. The recombination of benzyl radicals with allyl radicals produced 4-phenyl-1-butene, which subsequently are converted to dihydronaphthalene, and eventually to naphthalene. Considering the similarity between benzyl and allyl radicals, the rate for recombination of benzyl radicals was used to model recombination of benzyl and allylic radicals.

Another important recombination of benzyl radical occurred with the propargyl radical. Matsugi and Miyoshi [43] studied this reaction theoretically using CBS-QB3, B3LYP, and CASPT2 methods, and suggested that this reaction could be an important source of naphthalene, or methyleneindene formation. They suggested that the reaction of benzyl radical with propargyl radicals at low temperatures results in the production of stabilized 3-butylnylbenzene, or 2,3-butadienylbenzene adducts [43]. The adducts can undergo unimolecular isomerization through formation of diradical intermediates, leading to production of methylene-indanyl radicals, which subsequently decompose to methylene indene. Matsugi and Miyoshi also suggested that at high temperatures, the recombination reaction through chemically-activated pathways, directly results in the production of methylene-indanyl radicals, skipping the formation of the adducts. Based on their findings [43], the current mechanism includes both the formation of the stabilized adducts

and the chemically-activated reaction from the benzyl+propargyl reaction.

As methyl radicals are found in significant amounts in flames and high temperature oxidation, the present mechanism considers the recombination reaction of benzyl radicals with methyl radicals. This reaction leads to the production of ethylbenzene; and the rate for this reaction has been adopted from the recent study of Matsugi and Miyoshi [44].

3.3. PAH growth through addition reactions with stable intermediates

Reactions of benzyl radical with acetylene have been demonstrated to be an important reaction channel leading to the production of indene [45–47]. This reaction system in the current study has been described in accordance with the findings of Mebel et al. [47], who studied the reaction pathways and computed the rate constants at G3(MP2,CC)/B3LYP/6–311G(d,p) level of theory. The findings of Mebel et al. [47] suggest that at low temperatures and high pressures, this reaction precedes to form a stabilized adduct, which could subsequently produce indene+H or 3-phenylpropyne+H. Of the two bimolecular product pairs, the formation of the former is generally dominant at temperatures less than 2000 K. At low pressure and high temperature conditions, (identical to those in the present experiments), where the stabilization of the adduct is not possible, indene+H is expected to be the dominant product of this addition reaction.

To our knowledge, the reaction of benzyl with other stable intermediates such as ethylene, propyne, allene or butadiene, has not been studied theoretically and their contribution to the formation of PAHs is not known. For these reactions, analogies were used—when available—to describe the reaction pathways and rate parameters. Analogies from the reaction of allyl radical with ethylene, from a study by Wang et al. [48], were used to describe reactions of benzyl with ethylene.

By comparing with other kinetic models, current model showed better prediction against the experimental data (details of models comparison can be found in Section 1 of SM).

4. Results and discussion

4.1. PRF pyrolysis in JSR

The pyrolysis of a binary *n*-heptane/*iso*-octane mixture was first investigated here to clarify the effect of linear/branched alkanes on PAH formation, and to validate the proposed kinetic model against simpler mixtures. To this end, the pyrolysis of primary reference fuel PRF 70 was conducted in the JSR at temperatures between 790 K and 1210 K. Fuel consumption as a function of temperature is shown in Fig. 1. While the start of conversion of both *n*-heptane and *iso*-octane was observed at similar temperatures (around 880 K), the complete conversion of *iso*-octane into smaller molecules was observed at 1090 K, while *n*-heptane was consumed completely at 1150 K. The start of consumption and relative rates of consumption of both *n*-heptane and *iso*-octane were well predicted by the kinetic mechanism. The rate of production (ROP) analysis at 950 K (details of the ROP can be found in Section 2 of MS) suggested that the main pathway for *n*-heptane consumption was from H-atom abstraction reactions by H and CH_3 producing *n*-heptyl radicals. The *n*-heptyl radicals subsequently decomposed into small species, such as ethylene (C_2H_4), propene (C_3H_6), etc. In the case of *iso*-octane, the major consumption pathways included the initiation reactions involving breakage of the C–C bonds, along with the H-abstraction reactions. The main products of PRF 70 pyrolysis were ethylene (C_2H_4), acetylene (C_2H_2), methane (CH_4), propylene (C_3H_6), allene (C_3H_4 -A) and propyne (C_3H_4 -P), as shown in Fig. 2. The computed concentration profiles of those species

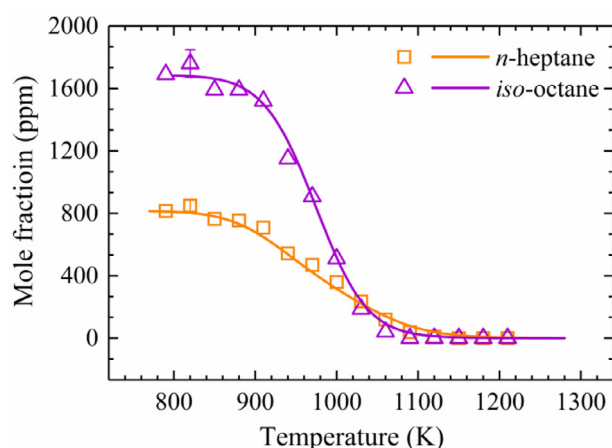


Fig. 1. Mole fraction profiles of fuel components (*n*-heptane/*iso*-octane) in JSR reactor pyrolysis of PRF 70 in 99.75% nitrogen at 1 atm and 1 s residence time. Symbols are experimental data; solid lines are simulated profiles.

also showed good agreement with the experimental results. From Fig. 2 it is clear that C_2H_4 , CH_4 , and C_3H_6 were formed above 880 K, while C_2H_2 , C_3H_4 -P and C_3H_4 -A were produced above 1000 K. C_2H_4 and C_3H_6 were produced from the decomposition of fuel radicals, while methane (CH_4) was produced from H-abstractions by methyl radical (CH_3) from the fuel molecules. C_2H_2 , C_3H_4 -A and C_3H_4 -P are seldom produced from decomposition of *n*-heptane and *iso*-octane, and are primarily produced from H-eliminations from vinyl (C_2H_3) and allyl (C_3H_5 -A) radicals derived from C_2H_4 and C_3H_6 . This explains the reason for the later production of C_2H_2 , C_3H_4 -A and C_3H_4 -P when compared to C_2H_4 and C_3H_6 .

In addition to small species, some mono- and di-aromatics were also observed in high temperature pyrolysis of PRF 70. The aromatics detected included benzene, toluene, ethylbenzene, styrene, indene, and naphthalene. Figure 3 shows the comparison of experimental and predicted concentrations of aromatics for the temperatures investigated in the present experiments. As seen in Fig. 3, the kinetic model successfully captured the concentration profiles of benzene, indene and naphthalene; but the kinetic model under-predicted concentrations of the other aromatics. In the experiments the production of benzene was observed to start around temperature of 960 K with concentration of benzene being close to 10 ppm, and the concentration of benzene was found to increase with increase in temperature. ROP analysis conducted with the mechanism suggests that for the temperatures of 900–1200 K, benzene was mainly produced from the elimination of H-atoms

from the cyclohexadienyl radical (CYC_6H_7), $CYC_6H_7 \rightleftharpoons C_6H_6 + H$, for example with 87% contribution at 1150 K shown in Fig. 16. Cyclohexadienyl radicals were produced from the ring expansion reaction of methyl-cyclopentadienyl radicals, which were the result of the recombination of reactions of methyl and cyclopentadienyl radicals. Cyclopentadiene (CPD), the precursor for cyclopentadienyl radical was found to be produced from 1-pentene chemistry and $C_2 + C_3$ reactions which shall be discussed in Section 4.5. The importance of 1-pentene chemistry towards CPD has been found to be important in recent pre-mixed flame study of 1-pentene [49].

Multiple reaction pathways contributed to the formation of toluene $C_6H_5CH_3$, including the reaction of phenyl (C_6H_5) with methyl ($C_6H_5 + CH_3 \rightarrow C_6H_5CH_3$ (+M)), the reaction of benzene with methyl ($C_6H_6 + CH_3 \rightleftharpoons C_6H_5CH_3 + H$). Ethylbenzene resulted from the recombination reaction of benzyl and methyl radicals ($C_6H_5CH_2 + CH_3 \rightleftharpoons C_6H_5C_2H_5$). Two main pathways contributed to the formation of styrene, including multi-step/direct dehydrogenation of ethylbenzene and allylidene-cyclopentadiene (C_5H_5CCVCH), which explained the higher concentration of styrene than of ethylbenzene. Allylidene-cyclopentadiene—a precursor to formation of styrene—was produced from the recombination reaction between cyclopentadienyl and propargyl radicals; this reaction has been modelled according the recommendations of Sharma et al. [50]. From the comparison shown in Fig. 3(a) it is clear that the mechanism underpredicts the concentrations of toluene, ethylbenzene, and styrene. Based on our analysis we believe the underprediction of ethylbenzene and styrene is result of underprediction of toluene, and revisiting of formation of toluene would benefit the PAH models. The recent work of Baroncelli et al. also highlighted the need for revisiting the kinetics of formation of toluene, and the C_8 aromatics [51].

In the experiments, indene and naphthalene were produced at higher temperatures than the monoaromatics, with detectable concentrations observed at around 1030 K. Indene and naphthalene are expected to be produced from the reactions of monoaromatics, clarifying the relative delay in their production compared to the monoaromatics. This dependence was also observed in kinetic simulations with the present PAH mechanism. According to the ROP at 1150 K, 43% of indene was produced from elimination of H-atoms from phenyl-allyl radical ($C_6H_5C_3H_4$), and 41% was from the reaction of benzyl with acetylene ($C_6H_5CH_2 + C_2H_2 \rightarrow IND + H$), and small amounts ~10% of indene was produced from unimolecular isomerization of phenylallene to indene ($C_6H_5CHCCH_2 \rightarrow IND$). Phenyl-allyl radical was produced by H-abstractions from phenylpropenes generated from recombination reactions of allyl and phenyl radicals. Similarly, phenylallene was produced from recombination reaction of phenyl radical with propargyl radical ($C_6H_5 + C_3H_3 \rightarrow C_6H_5CHCCH_2$). The kinetics of recombination of

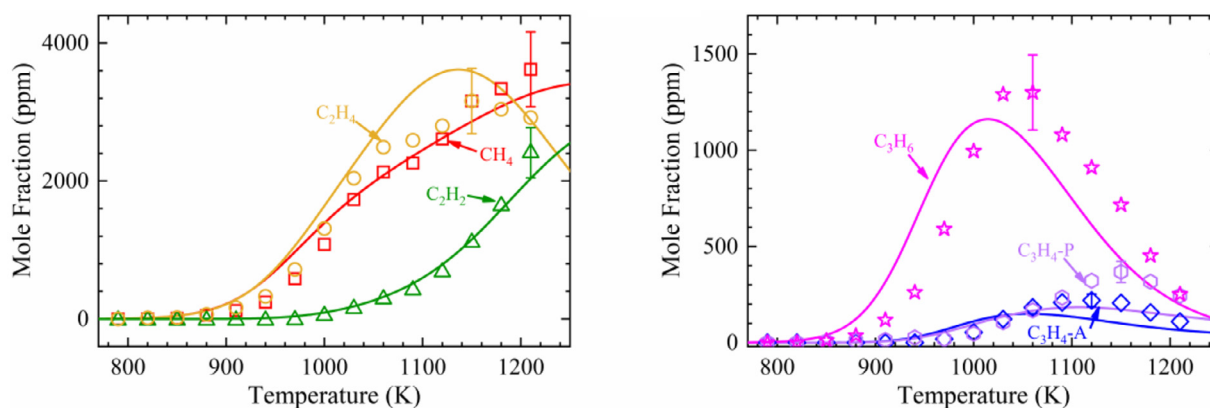


Fig. 2. Mole fraction profiles of small species in JSR pyrolysis of PRF 70. Symbols are experimental data; solid lines are simulated profiles.

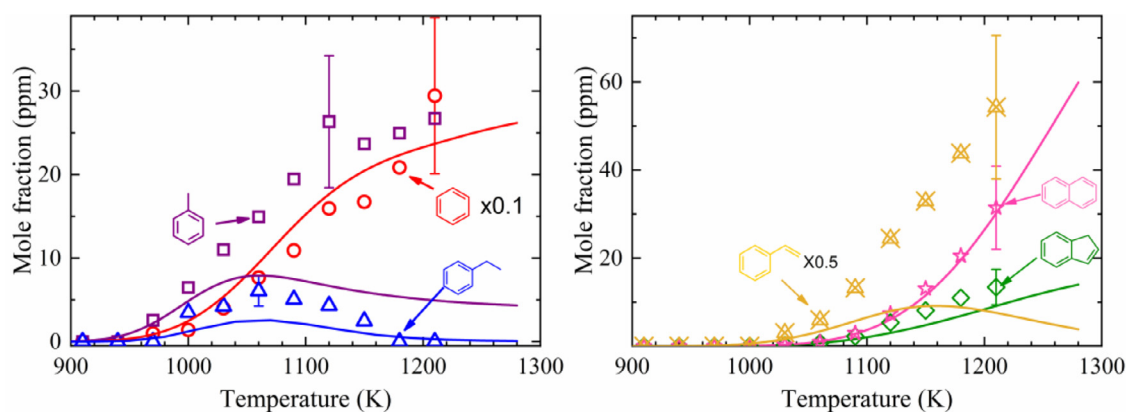


Fig. 3. Mole fraction profiles of benzene (C_6H_6), toluene ($C_6H_5CH_3$), ethylbenzene ($C_6H_5C_2H_5$), styrene ($C_6H_5C_2H_3$), indene (C_9H_8), and naphthalene ($C_{10}H_8$) in JSR pyrolysis of PRF 70. Symbols are experimental data; solid lines are simulated profiles.

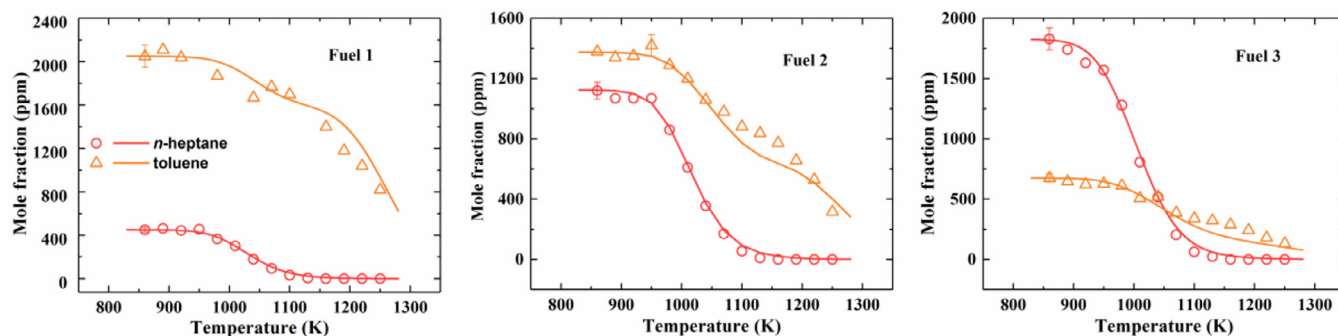


Fig. 4. Mole fraction profiles of fuel components (*n*-heptane/toluene) in JSR reactor pyrolysis of Fuels 1, 2 and 3. The amount of *n*-heptane increases and the toluene decreases in the frames from left to right. Symbols are experimental data; solid lines are simulated profiles.

phenyl radical with propargyl and subsequent isomerization of has been modelled according to findings of recent work of Morozov and Mebel [52]. For naphthalene, three main paths contributed to its formation, including the methyl assisted ring expansion reaction of indenyl radical, the dehydrogenation reaction of dialin radicals, and the recombination reaction of C_7H_5 with C_3H_3 ($C_7H_5 + C_3H_3 \Rightarrow NAPH$). Further details of naphthalene reaction pathways will be discussed in Section 4.5.

4.2. Toluene/*n*-heptane blends pyrolysis in JSR

Binary *n*-heptane/toluene mixtures were studied next to understand the role of aromatics in gasolines on PAH formation. The pyrolysis of three *n*-heptane/toluene mixtures was studied in the JSR over a temperature range of 860 K to 1250 K. The concentration profiles of the fuel molecules from both experiments and simulations are compared in Fig. 4. As seen in the figure, the kinetic model captured very well the conversion of parent fuel molecules as the amount of *n*-heptane is decreased and the toluene is increased in the results from the left frame to the right frame. The conversion of both fuels began at around 960 K for fuel blends 1 and 2, while conversion started at lower temperatures around 900 K for Fuel 3. The kinetic model also captured characteristic differences among the fuel blends. For these blends, the main pathway of *n*-heptane consumption was a unimolecular decomposition reaction to *n*-propyl (NC_3H_7) and *n*-butyl (PC_4H_9) radicals ($NC_7H_{16} = PC_4H_9 + NC_3H_7$). The conversion of toluene also began at around 960 K, as shown in Fig. 4, but the rate of consumption was much lower than the *n*-heptane. For Fuels 1 and 2, the rate of toluene consumption increased at 1150 K. According to the rate of production (ROP) analysis, at temperatures above 1050 K, the main pathways of toluene consumption were the

hydrogen abstraction reactions ($C_6H_5CH_3 + CH_3 \rightleftharpoons C_6H_5CH_2 + CH_4$, $C_6H_5CH_3 + H \rightleftharpoons C_6H_5CH_2 + H_2$) and the formation of benzene via the ipso addition reaction with methyl radical ($C_6H_5CH_3 + H \rightleftharpoons C_6H_6 + CH_3$).

Figure 5 shows the concentration profiles of small species quantified in JSR pyrolysis of *n*-heptane/toluene mixtures, which include methane, ethylene, acetylene, allene, propyne and propylene. At the temperatures studied here, the decomposition of the aromatic ring was highly unlikely; and all the small species resulted primarily from the decomposition of *n*-heptane. This observation was corroborated by the fact that the concentration of all the small species was higher for Fuel 3, which consisted of higher amounts of *n*-heptane. This tendency was well predicted by the proposed kinetic model. While the experimental and predicted mole fractions of ethylene, methane, and acetylene match closely, significant differences were observed in the concentrations of allene, propyne and propene. The mechanism underpredicted the concentrations of propene at temperatures above 1025 K and this also impacted the concentrations of allene and propyne which were derived by allyl radical produced from propene. So, the observed differences in propene, allene and propyne concentrations could be attributed to the shortcoming of the mechanism in predicting the produced concentration of propene and also the consumption of propene. Furthermore, the concentrations of propyne in experiments exhibited a bimodal nature which was not observed in simulations and could be an experimental artifact.

The concentration profiles of benzene, styrene and ethylbenzene are shown in Fig. 6. The trends observed in benzene were similar in the experiments and simulations. The formation of benzene began around 950 K and coincided with the temperature at which conversion of toluene was observed. Between 1050 K and 1200 K, the concentration of benzene in Fuel 2 pyrolysis was

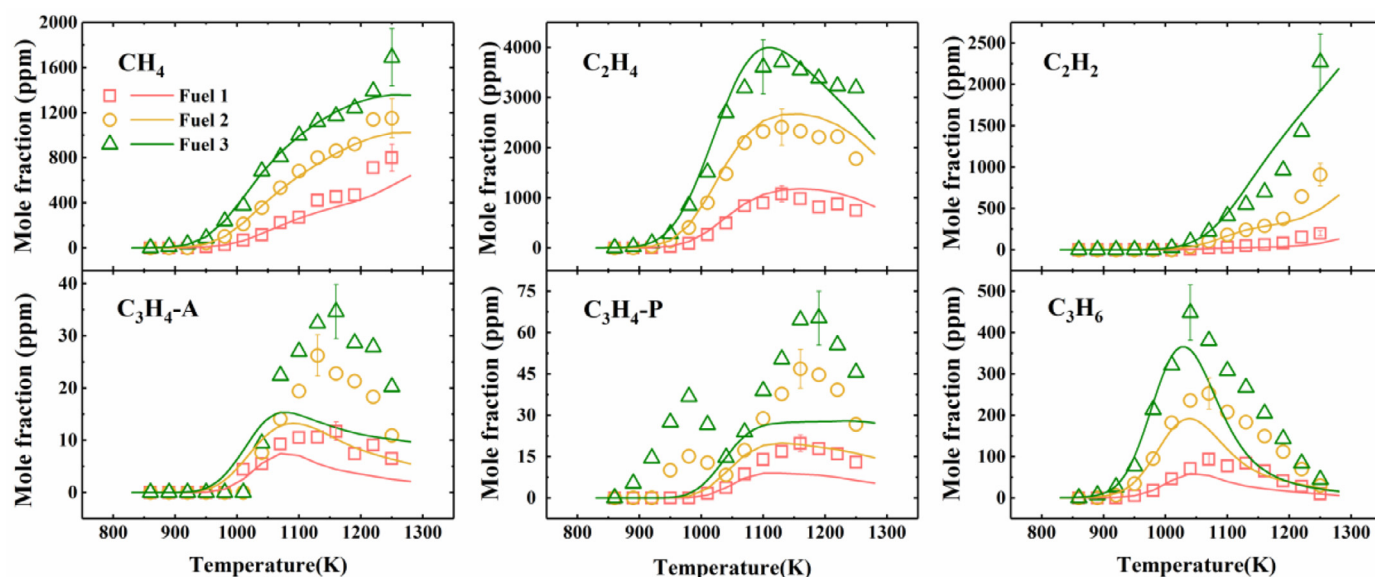


Fig. 5. Mole fraction profiles of small species in JSR pyrolysis of Fuels 1, 2 and 3. Symbols are experimental data; solid lines are simulated profiles.

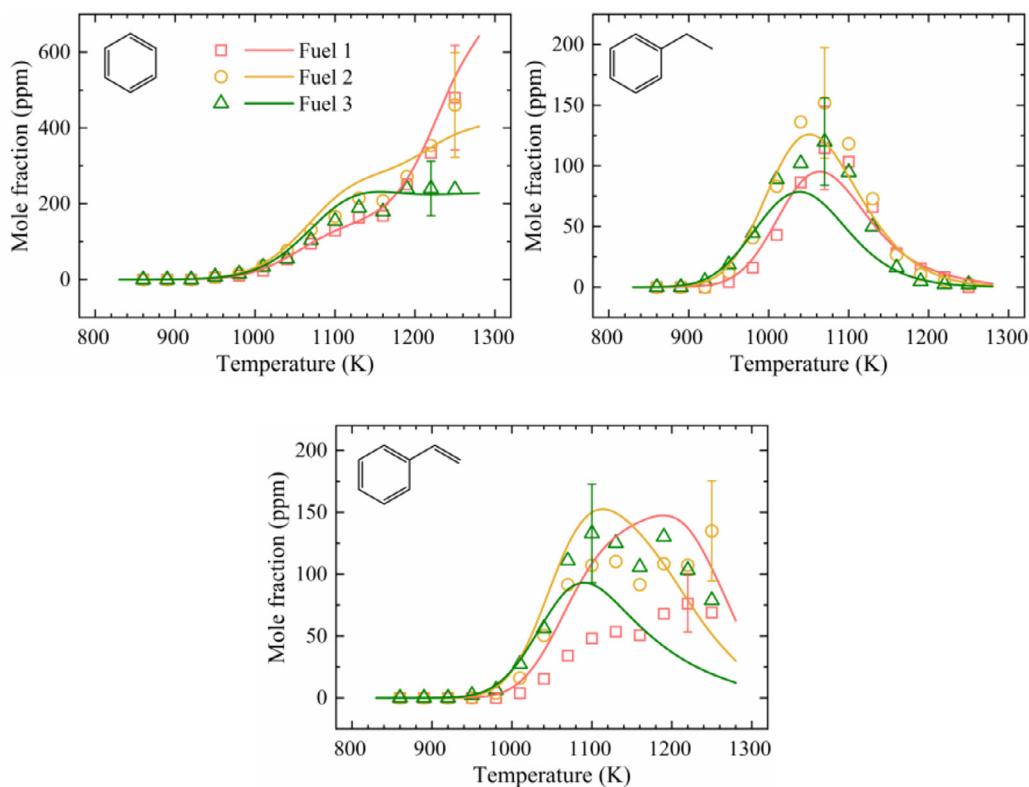


Fig. 6. Mole fraction profiles of benzene (C_6H_6), ethylbenzene ($C_6H_5C_2H_5$) and styrene ($C_6H_5C_2H_3$) in JSR pyrolysis of Fuels 1, 2 and 3. Symbols are experimental data; solid lines are simulated profiles.

marginally higher compared to the other two fuel blends. Interestingly, Fuel 1 produced the least amount of benzene below 1180 K, even though it contained the highest amount of toluene. According to ROP analysis, at 1150 K benzene was mainly produced from toluene through the reaction $C_6H_5CH_3 + H = C_6H_6 + CH_3$ in all the *n*-heptane/toluene mixtures. However, the dehydrogenation reaction of cyclohexadienyl radical (CYC_6H_7), $CYC_6H_7 = C_6H_6 + H$ also contributed 2% and 18% to benzene production in Fuels 2 and 3, respectively. CYC_6H_7 was produced from reactions initiated by

a recombination reaction of methyl and cyclopentadienyl radicals, both of which were produced from intermediates produced from unimolecular decomposition reactions of *n*-heptane. Above 1200 K, the generation rate of benzene reduced in Fuels 2 and 3, while it continued to increase in Fuel 1. This was caused by the complete consumption of *n*-heptane at higher temperatures, following which toluene became the only source of benzene production in this temperature region. For ethylbenzene ($C_6H_5C_2H_5$), all three *n*-heptane/toluene mixtures showed a similar tendency.

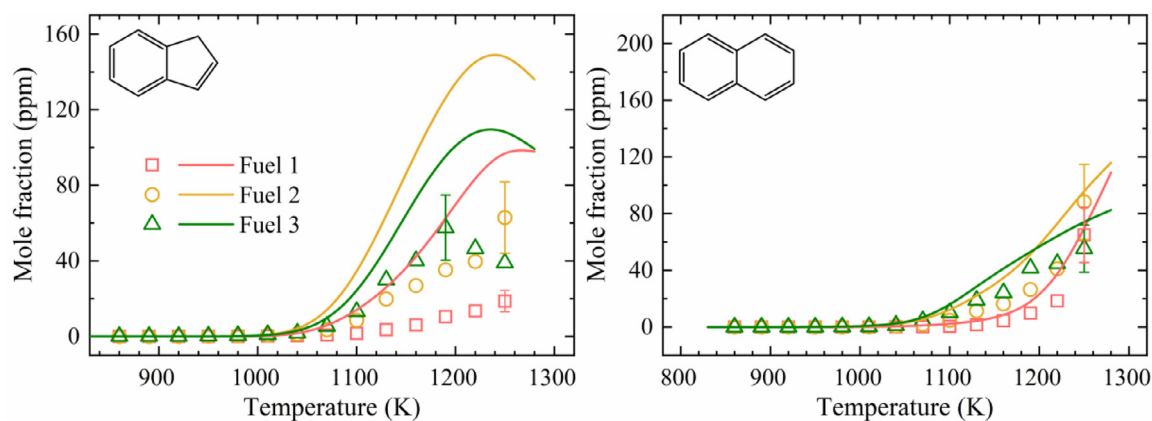


Fig. 7. Mole fraction profiles of (a) indene (C_9H_8) and (b) naphthalene ($C_{10}H_8$) in JSR pyrolysis of Fuels 1, 2 and 3. Symbols are experimental data; solid lines are simulated profiles.

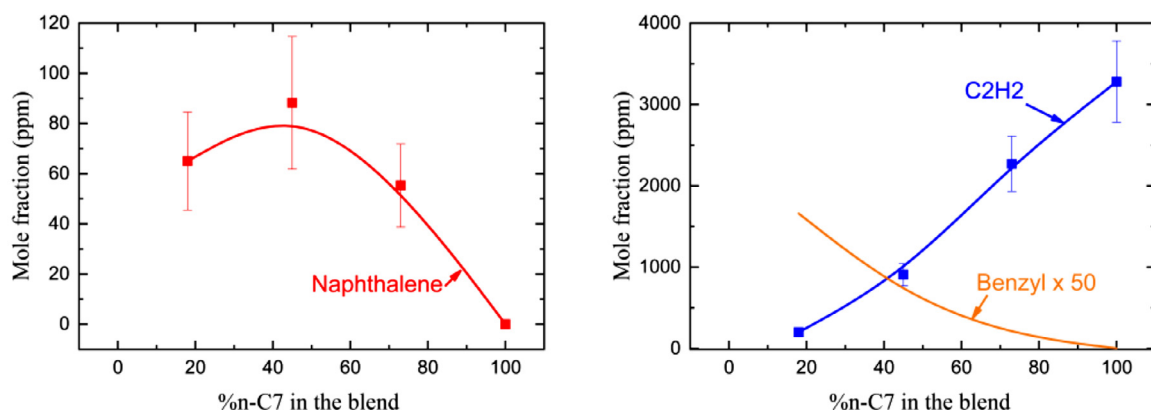


Fig. 8. Maximum concentration of naphthalene, acetylene and the benzyl formed in toluene/*n*-heptane mixtures. For naphthalene and acetylene, symbols are experimental data and lines are for guiding the eyes. For benzyl, the line is the calculated result.

From 950 K to 1050 K, ethylbenzene was produced by the reaction of the benzyl radical with methyl ($C_6H_5CH_2 + CH_3 = C_6H_5C_2H_3$), and all the benzyl radical was produced from H-abstractions from the methyl site in toluene. Above 1050 K, styrene ($C_6H_5C_2H_3$) was produced from the dehydrogenation reaction of ethylbenzene ($C_6H_5C_2H_3 = C_6H_5C_2H_2 + H_2$), which resulted in decreased concentrations of ethylbenzene.

Figure 7 shows the concentration profiles of indene (C_9H_8) and naphthalene ($C_{10}H_8$) in JSR pyrolysis of these three *n*-heptane/toluene mixtures. The experimental data indicates that Fuel 1 produced less indene and naphthalene than the other two blends; and Fuels 2 and 3 produced similar amounts of indene and naphthalene. The mechanism predicted the concentrations of naphthalene adequately for the three blends. For indene, the prediction for Fuel 3 was reasonable; however, for Fuels 2 and 3, the predicted increase in indene with temperature was too large. ROP analysis at 1150 K indicated that the main pathway of indene formation was from the reaction of the benzyl radical with acetylene ($C_6H_5CH_2 + C_2H_2 = C_9H_8 + H$) in all the *n*-heptane/toluene mixtures. It was apparent that the amount of indene produced depended on amounts of both the benzyl radical and acetylene, which explained why Fuel 1 (with high amounts of toluene) did not necessarily produce more indene. This indene formation, dependent upon concentrations of both acetylene and the benzyl radical, further verified the non-linear dependent indene formation with respect to toluene concentration and the synergistic effect between heptane and toluene. A similar synergistic effect can also be observed in the naphthalene profiles, in which Fuel 1 produced minimal naphthalene, although it contained more toluene than the other two

mixtures, as shown in Fig. 8. This synergistic effect of heptane addition on the formation of naphthalene was successfully captured by the kinetic model. The ROP analysis suggested that naphthalene was produced from reactions of benzyl with acetylene, or propargyl radical. The *in-situ* concentrations of benzyl radical were expected to be higher for blends containing high amounts of toluene; but the concentration of acetylene and propargyl (which are derived from the decomposition of heptane) were expected to be low for toluene-rich mixtures. Therefore, the blends with high amounts of either toluene or heptane may not be conducive to formation of naphthalene, and a local maxima might be observed for intermediate mixtures, which explains the synergistic behavior observed in Fig. 8.

4.3. TPRF mixtures pyrolysis in JSR

Finally, ternary mixtures of *n*-heptane/*iso*-octane/toluene were studied to understand PAH formation in fully formulated gasoline surrogate fuels. The pyrolysis of four TPRF mixtures with RON from 70 to 97.5, named TPRF70, TPRF 80, TPRF 91 and TPRF 97.5, are discussed in this section. The proportions of *n*-heptane, toluene, and *iso*-octane in each mixture can be found in Table 1. As previously mentioned, the experimental data of TPRF 70 and TPRF 97.5 are from [28]. Figure 9 shows the comparison of experimental and simulated mole fraction profiles of the heptane, *iso*-octane and toluene during pyrolysis of the four different TPRF mixtures. As seen in the figure, the prediction from the mechanism matched experimental results, with differences generally observed at temperatures over 1100 K, especially for the conversion profiles of toluene.

Table 1

Molar composition of gasoline surrogates and facilities used.

	<i>n</i> -Heptane	<i>iso</i> -Octane	Toluene	RON	Facility	Volume of JSR (cm ³)
PRF 70	0.33	0.67	0	70	JSR	31.6
Fuel 1	0.18	0	0.82	95	JSR	31.6
Fuel 2	0.45	0	0.55	63.2	JSR	31.6
Fuel 3	0.73	0	0.27	30	JSR	31.6
TPRF 70*	0.35	0.37	0.29	70	JSR & FR	76
TPRF 80	0.27	0.33	0.4	80	JSR	76
TPRF 91	0.17	0.29	0.54	91	JSR	76
TPRF 97.5*	0.15	0.08	0.78	97.5	JSR & FR	76
Pure Toluene*	0	0	1	120	FR	–

* Literature data [28,30].

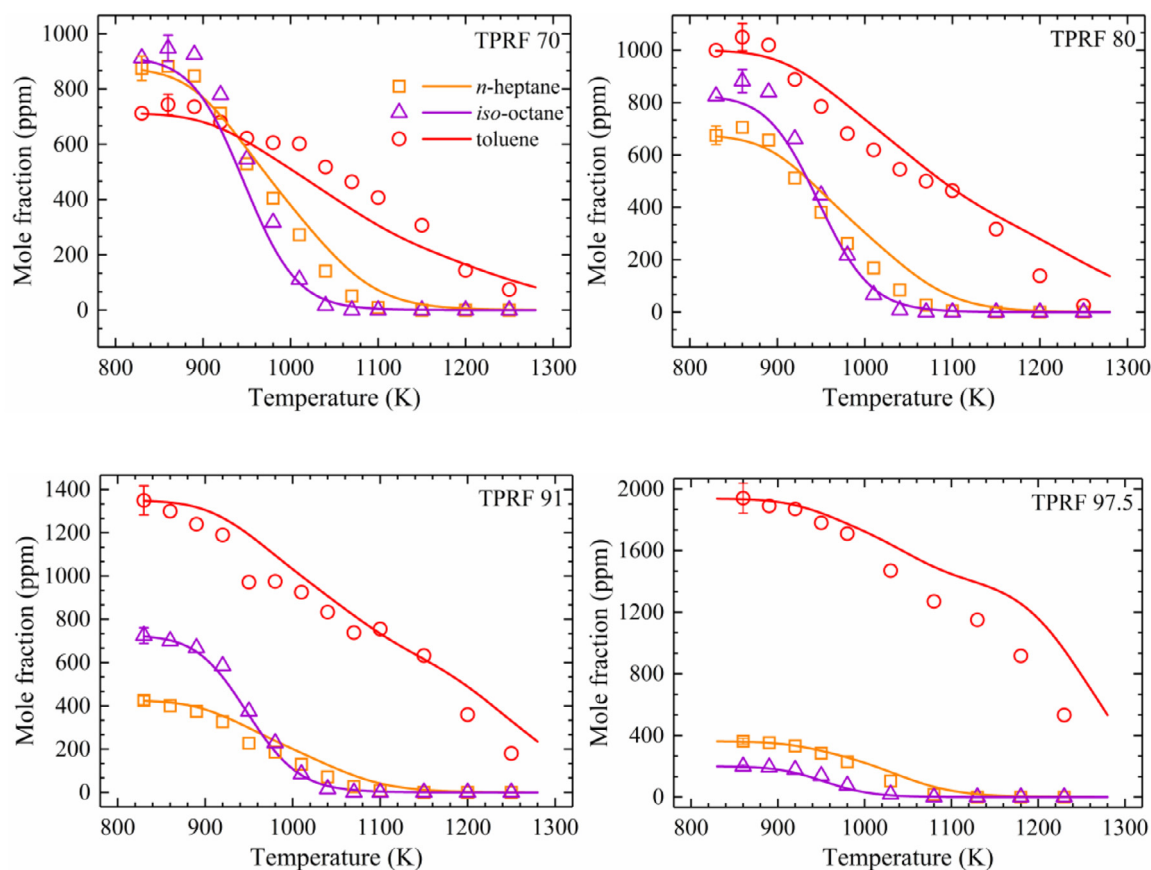
**Fig. 9.** Mole fraction profiles of fuel components (*n*-heptane/*iso*-octane/toluene) in JSR reactor pyrolysis of TPRF mixtures. Symbols are experimental data; solid lines are simulated profiles.

Figure 10 shows the mole fraction profiles of small species in the JSR pyrolysis of TPRF mixtures. Like *n*-heptane/toluene blends, almost all the small species were produced primarily from the decomposition of *n*-heptane and *iso*-octane. Concentrations of the small species were ranked as TPRF70 > TPRF80 > TPRF91 > TPRF97.5, consistent with the total amount of *n*-heptane and *iso*-octane within the mixtures. This trend was also predicted by the kinetic model. The mole fractions of small species were reasonably well predicted except for allene (C_3H_4 -A) which was over-predicted by a factor of 2–4.

Figures 11 and 12 show the comparison of aromatic formation in the pyrolysis process of TPRF70, TPRF80, TPRF91 and TPRF97.5. Note that the mole fractions of benzene are comparable in TPRF mixtures in the temperature range between 950 K and 1200 K. In TPRF mixtures, the main source for benzene formation is toluene, through the reaction $C_6H_5CH_3 + H = C_6H_6 + CH_3$. However, reactions $CYC_6H_7 = C_6H_6 + H$ and $2C_3H_3 = C_6H_6$ also contributed to the ben-

zene formation. For instance, in TPRF80 at 1150 K, the contribution rate of the pathway $C_6H_5CH_3 + H = C_6H_6 + CH_3$, $CYC_6H_7 = C_6H_6 + H$, and $2C_3H_3 = C_6H_6$ were 53%, 23%, and 2%, respectively. As noted in the *n*-heptane/toluene mixture, CYC_6H_7 was produced from recombination reactions between methyl and the cyclopentadienyl radical (C_5H_5). The main pathway for formation of C_5H_5 was H-abstraction from cyclopentadiene which was produced from reactions between allyl radical and acetylene.

It is clear that TPRF mixtures produced much more indene than PRF 70 in both experiments and simulations. For comparison, the peak concentration of indene (IND) during pyrolysis of PRF70 was less than 20 ppm, while the peak concentration of indene during pyrolysis of TPRF's was about 60 ppm. The kinetic model predicted the experimental trends for indene, but the peak mole fractions were shifted to higher temperatures by about 60 K. The main pathway (about 60% at 1180 K) of indene formation in PRF 70 was from the dehydrogenation reaction of the phenylallyl radical

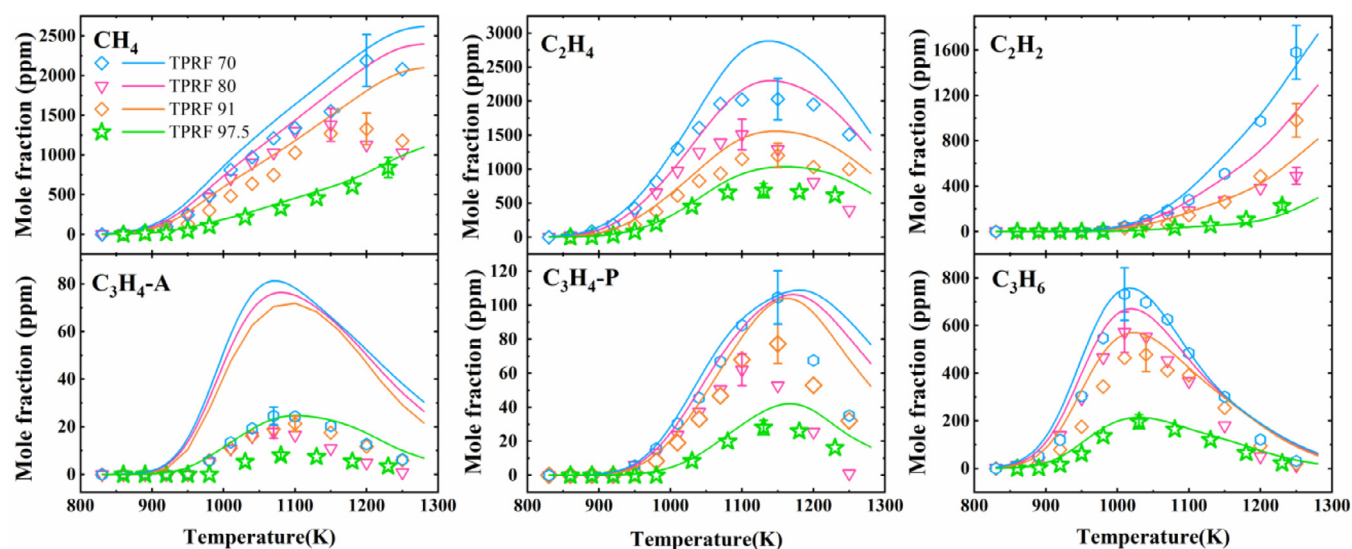


Fig. 10. Mole fraction profiles of small species in JSR pyrolysis of TPRF mixtures. Symbols are experimental data; solid lines are simulated profiles.

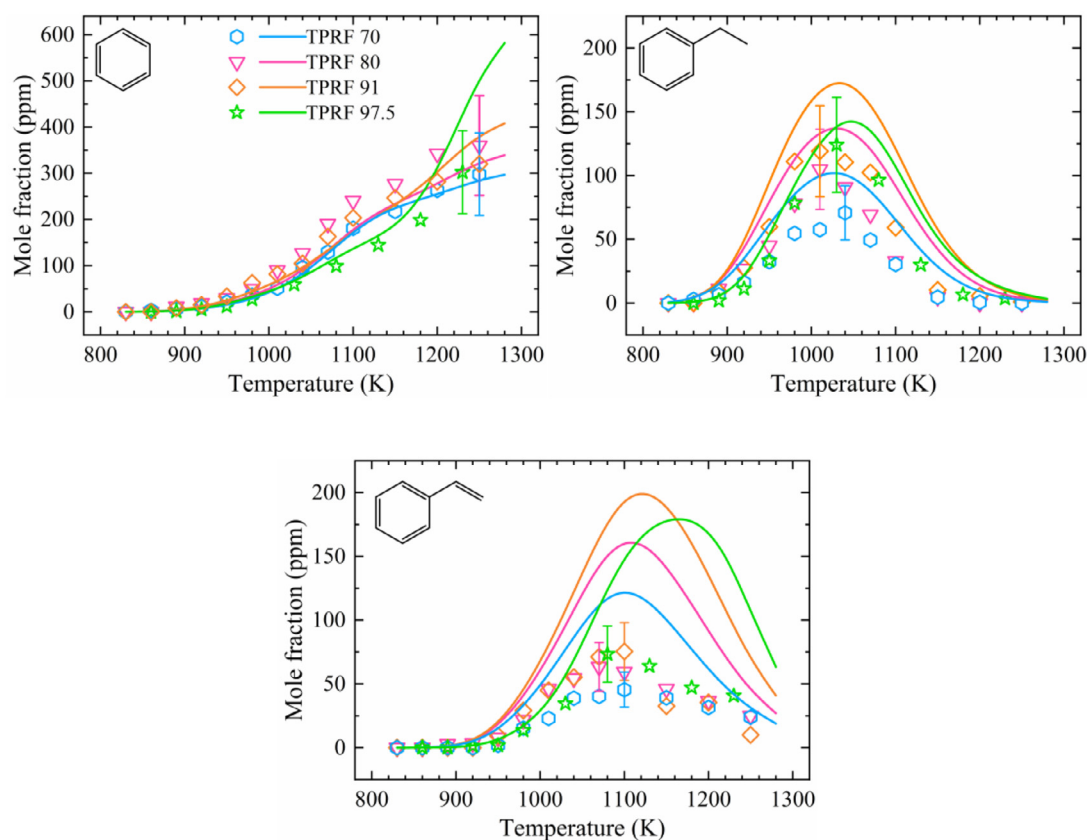


Fig. 11. Mole fraction profiles of benzene (C_6H_6), ethylene ($C_6H_5C_2H_5$) and styrene ($C_6H_5C_2H_3$) in JSR pyrolysis of TPRF mixtures. Symbols are experimental data; solid lines are simulated profiles.

($C_6H_5C_3H_4$, $C_6H_5C_3H_4=IND+H$). Another pathway of the indene formation was the reaction of the benzyl radical with acetylene ($C_6H_5CH_2+C_2H_2=IND+H$). For TPRFs, the main source of indene formation was the benzyl radical via $C_6H_5CH_2+C_2H_2=IND+H$. Concentrations of the indene in TPRFs were ranked as follows: $TPRF97.5>TPRF91>TPRF80>TPRF70$, which is the same order of the toluene containing ratio. Like the indene results, TPRFs produced much more naphthalene ($C_{10}H_8$) than PRF 70. The kinetic model also predicted this tendency accurately. For TPRF's the formation

of naphthalene was predominantly through 1-methylene-2-indanyl radical. The 1-methylene-2-indanyl radical could be produced from methyl+indenyl radical reaction and shall be discussed in Section 4.5.

4.4. Validation of larger PAHs in flow reactor

In this section, the mechanism has been validated against flow reactor data from the literature to test the fidelity of the mech-

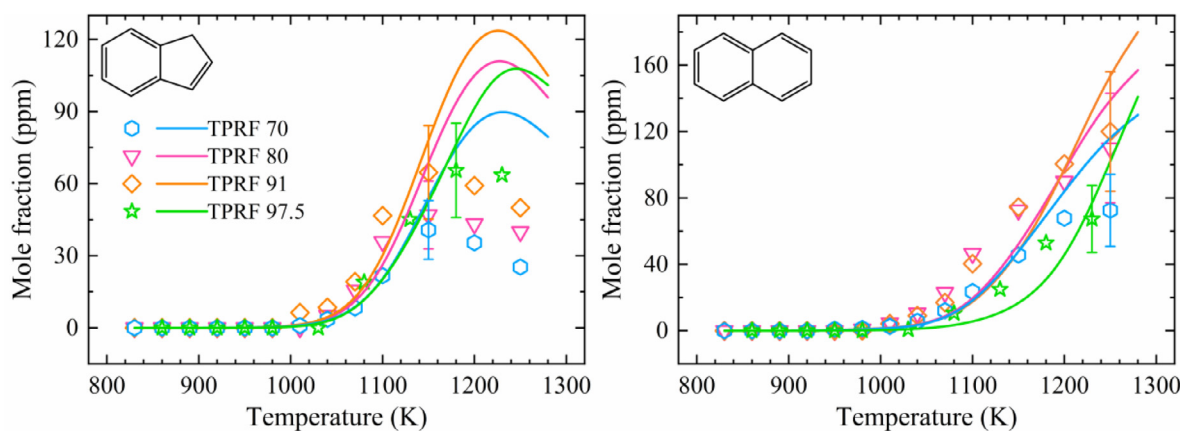


Fig. 12. Mole fraction profiles of (a) indene (C_9H_8) and (b) naphthalene ($C_{10}H_8$) in JSR pyrolysis of TPRF mixtures. Symbols are experimental data; solid lines are simulated profiles.

anism in capturing the formation of tri- and tetra-ring aromatics during gasoline surrogate pyrolysis [28,30]. Figure 13 shows the mole fraction profiles of phenanthrene and pyrene isomers in flow reactor pyrolysis of TPRF70 and 97.5. Overall, the current model predicted the phenanthrene well in both TPRF 70 and 97.5, with differences well within the experimental uncertainty (factor of two). The main pathways for formation of phenanthrene in TPRF fuels were the re-combination reaction of the fulvenallenyl radical ($2C_7H_5 \Rightarrow PHNTHRN$), reaction of phenylacetylene with phenyl radical ($C_6H_5C_2H + C_6H_5 \rightarrow PHNTHRN$), recombination of benzyl radical, and methyl-assisted ring expansion of fluorene radical. Figure 13(c) shows the reaction network for formation of phenanthrene in pyrolysis of TPRF 70 for inlet temperature of 1343 K.

As stated in Section 3.1, the self-recombination reaction of the fulvenallenyl radical was modelled according to the suggestions of da Silva and Bozzelli [16]; this channel contributes to approximately 28% of the production of phenanthrene. The reaction of phenyl radical with phenylacetylene was modelled based on the recommendations of Aguilera-Iparraguirre et al. [42], and this reaction contributes to approximately 28% of the flux of phenanthrene formation. One of the key steps in formation of phenanthrene from this reaction is formation of *-ortho* 1,2- diphenylethylene radical (*Int#1*, in Fig. 13(c)). This radical can be produced from reaction network initiated by recombination of benzyl radicals leading to formation of 1,2-diphenylethane. 1,2- Diphenylethane, through dehydrogenation reactions, produces 1,2- diphenylethylene that upon H-abstraction from ring produces *Int#1*. This reaction network initiated by self-recombination of benzyl radicals effectively contributes to about 15% of formation of phenanthrene for the condition analyzed. Formation of phenanthrene from fluorene radicals has also been found to be important, and is consistent with the findings of [30,44]. This ring expansion reaction, in principle, is similar to formation of benzene and naphthalene from cyclopentadienyl and indenyl radicals, respectively. Fluorene in the present simulations was found to be produced from reactions of benzyl radical with *o*-benzyne, and has been modelled according to the recommendations of Matsugi and Miyoshi [36].

Regarding pyrene isomers (pyrene, acephenanthrylenes, fluoanthrene, ethynyl-phenanthrene), the model under-predicted the peak concentration observed in experiments by about a factor of two, which is close to reported uncertainty in measurements. In the current mechanism, pyrene isomers are predominantly produced from phenanthrene via the classic HACA (hydrogen abstraction reaction acetylene addition) scheme as shown in Fig. 13(d).

The model was also validated against the toluene pyrolysis experiment in the flow reactor, comparison of the experiment and

simulated results are demonstrated in Fig. 14. Simulated benzene profiles agreed well with experiments, as did simulated phenanthrene profiles below 1200 K. The reaction pathways for production of phenanthrene are similar to those shown in Fig. 13(c). The predicted indene mole fractions agreed reasonably with the experimental data below 1250 K. However, naphthalene mole fractions were underpredicted. Like TPRFs, the current model underestimated the formation of pyrene in toluene pyrolysis. The main pathways of pyrene isomers mentioned earlier in the TPRF pyrolysis are also important in toluene pyrolysis.

4.5. ROP analysis of benzene and naphthalene formation in various gasoline surrogates

Kinetic analyses were conducted to identify the important reaction pathways controlling the formation of benzene and naphthalene during pyrolysis of PRF 70, Fuel 1, TPRF 80 and TPRF 97.5 surrogates, and to clarify the changing importance of different reaction pathways to the change in surrogates' composition. We began with kinetic analyses of the formation of benzene and later, focused on the formation of naphthalene.

Identification of the top six reaction pathways for formation of benzene and its isomers (fulvene and 2-ethynyl 1–3-butadiene) was based on rate of production analyses based on the current mechanism. Figure 15 shows the top six reactions identified from ROP analysis and the relative contribution of each reaction is shown in Fig. 16. The first pathway is the ring formation of C_6H_6 isomers via the recombination reaction of allyl (C_3H_5-A) and propargyl radical (C_3H_3). The second pathway is the self-recombination propargyl radicals (C_3H_3). Third is from the toluene reaction ($C_6H_5CH_3 + H = C_6H_6 + CH_3$). Pathway number four is from recombination reactions cyclopentadienyl and methyl radicals. The fifth and sixth pathways are quite similar, in which benzene is produced *ipso*- substitution reactions of styrene and ethylbenzene. Reaction pathways 1–4, have been identified as important for formation of benzene isomers in aliphatic and aromatic fuels from earlier studies too. However, the importance of these pathways in gasoline surrogates is still not clear, and so the current analyses can be helpful to understand the formation pathways for benzene (and also for naphthalene). All the rates of production for the six pathways in PRF 70, Fuel 2, TPRF80, and TPRF 97.5 at 1150 K, are shown in Fig. 16. In Fig. 16, the % shown reflect the importance of the particular reaction towards formation of benzene isomers. As seen in Fig. 16, pathway # 3 (*ipso*- displacement reaction: $C_6H_5CH_3 + H = C_6H_6 + CH_3$) dominated the benzene for-

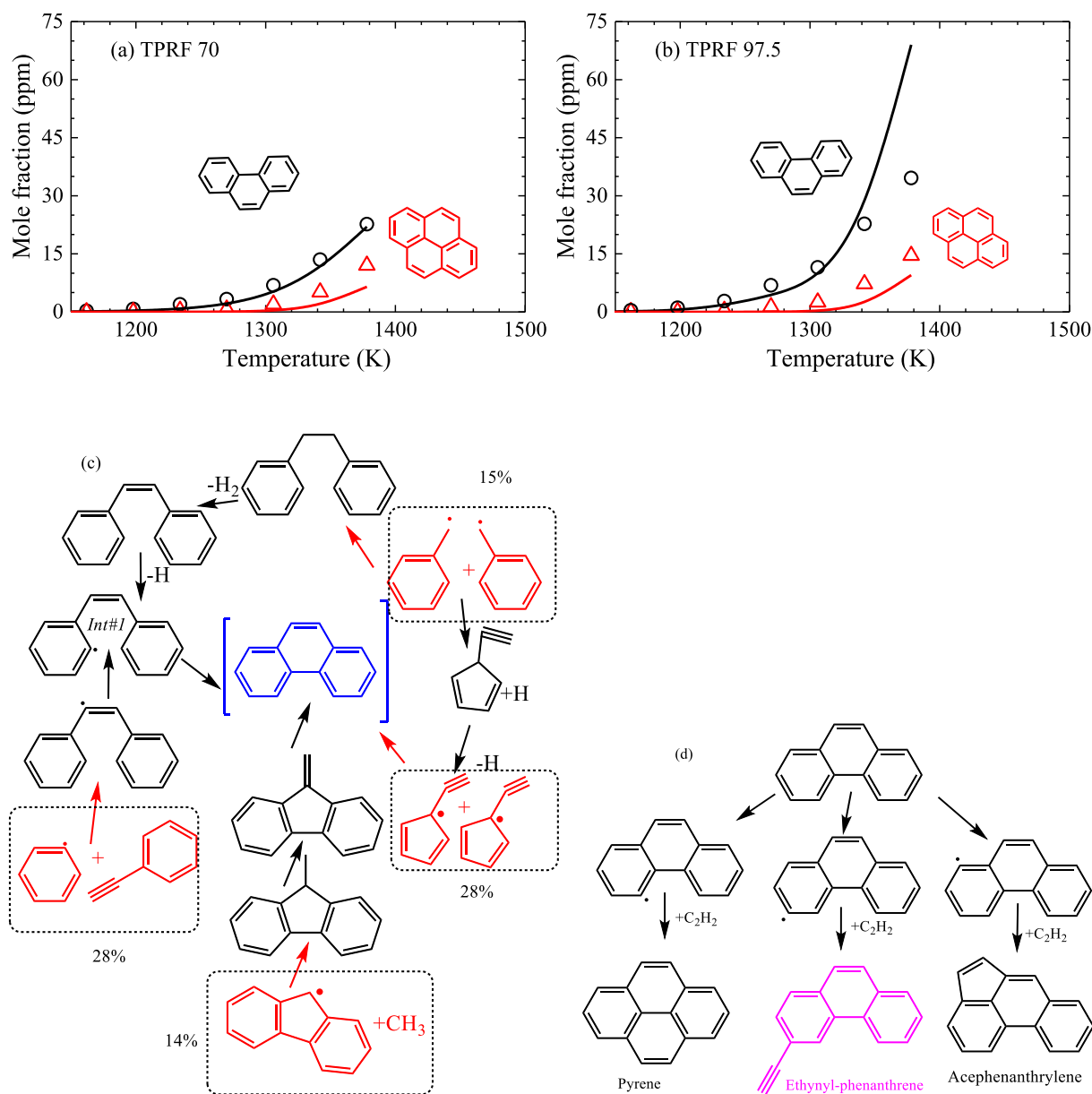


Fig. 13. (a) & (b) Mole fraction profiles of phenanthrene ($C_{14}H_{10}$) and pyrene isomers ($C_{16}H_{10}$) in flow reactor pyrolysis of TPRF 70 and 97.5, respectively. (c) and (d) Important reaction pathways for formation of phenanthrene ($C_{14}H_{10}$) and pyrene isomers ($C_{16}H_{10}$) as predicted by the mechanism. Symbols are experimental data taken from [28]; solid lines are simulations by current model.

mation in Fuel 2, TPRF80, and TPRF 97.5, all of which contained toluene. Since PRF 70 does not contain toluene as a base fuel, and it only produced toluene concentrations less than 30 ppm from Fig. 3, the third pathway did not contribute much to benzene formation for PRF 70. For PRF fuels, decomposition of *n*-heptane and *iso*-octane produce relatively high concentrations of C_2H_2 , C_3H_4 -P and C_3H_4 -A, and the pathways related to those radicals become dominant for benzene. For conditions investigated herein, formation of benzene during pyrolysis of PRF 70 was found to be majorly from reaction pathway #4. In our analysis we found cyclopentadienyl radical (C_5H_5) to be produced from cyclopentadiene (C_5H_6), which is produced primarily from C_2+C_3 chemistry as shown in Fig. 17. These C_2+C_3 reactions in the current mechanism have been described based on the findings of Wang et al. [46]. We also find the 1-pentenyl radicals to contribute to formation of cyclopentadiene with the importance of these reactions decreasing with increase in temperature for the conditions investigated here in the JSR. Reac-

tion network #4 was found to be important for TPRF fuels too, and contributing to about 23% of the benzene formation during pyrolysis of TPRF 80. Among the two TPRF fuels analyzed in Fig. 16, we find pathway #4 to be more important towards benzene formation for TPRF 80 than in the case of TPRF 97.5. This can be attributed to that fact that TPRF 80 has more paraffinic content than in TPRF 97.5. It is to be noted that the aromatic content in market gasolines is generally about 25–35%, which is close to the toluene content in TPRF 80. Thus, it is expected that the contribution of pathways #4 and #3 towards formation of first aromatic ring during pyrolysis of market gasoline fuels are comparable. Moreover, pathway #5 also made a few contributions of benzene formation in toluene-containing fuels, Fuel 2, TPRF 80, and TPRF 97.5.

Four pathways of naphthalene are also shown in Fig. 18, including (1) ring expansion reactions of methylindene radicals which produced from indenyl radical with methyl radical; (2) ring expansion reactions of 1-methylene-2-indanyl radicals which produced

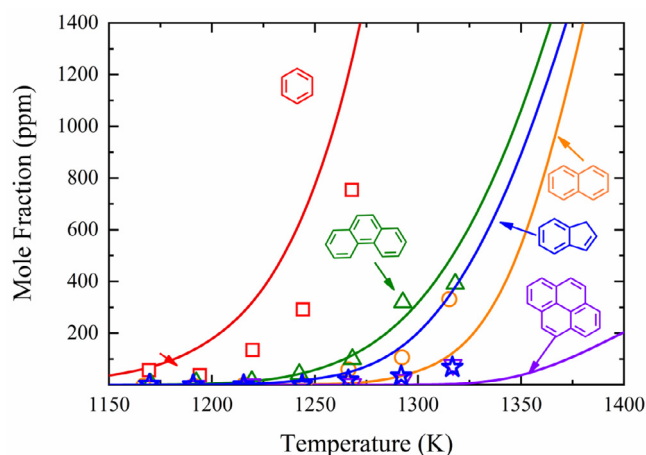


Fig. 14. Mole fraction profiles of benzene (C_6H_6), naphthalene ($C_{10}H_8$), phenanthrene ($C_{14}H_{10}$) and pyrene ($C_{16}H_{10}$) in flow reactor pyrolysis of toluene. Symbols are experimental data taken from [30]; solid lines are simulated from the current model.

from benzyl ($C_6H_5CH_2$) with propargyl (C_3H_3); (3) reaction of fulvenallenyl radical (C_7H_5) with propargyl (C_3H_3); (4) dehydrogenation of reaction of dialin radical ($C_{10}H_9$).

Figure 19 shows relative contribution of these five pathways to the production of naphthalene for PRF 70, Fuel 2, TPRF80, and TPRF 97.5 at 1150 K. Clearly, the methylindene radical pathway was dominant in the naphthalene formation for all four selected gasoline surrogates with the methylindene primarily produced from indenyl+CH₃ radical. This reaction in the current mechanism has been modeled according to recent study from Zhao et al. [53] which focuses on indenyl+CH₃ radical. For PRF 70, the third and fourth paths contributed the remaining 35.05% and 10% of naphthalene formation, but their contributed more in not significant the other three fuels. The main reason is that these two paths required higher concentrations of small species like C_2H_2 and C_3H_4 -P to proceed. Reaction network #2, the recombination reaction between propargyl and benzyl radical, also contributes to formation of naphthalene with its importance being the highest for TPRF 97.5. This increase in importance of Reaction network #2 for TPRF 97.5 can be attributed to the slower build of acetylene. From Fig. 10 it can be seen in the case of TPRF 97.5, the production of acetylene is very weak at temperatures below 1200 K, while concentrations of allene and propyne which are precursors for propargyl already peak by 1200 K. As a result, the flux through recombination reaction of benzyl with propargyl is

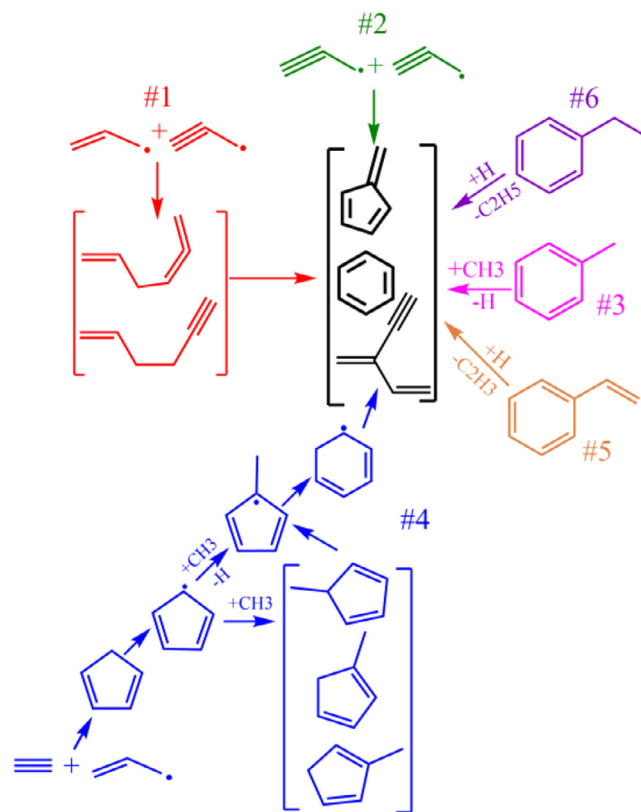


Fig. 15. Pathways to the formation of benzene and its isomers: fulvene and 2-ethynyl 1-3-butadiene at 1150 K.

higher for the case of TPRF 97.5. This analysis clearly demonstrates the importance of acetylene, propargyl and benzyl radicals to naphthalene formation in all gasoline fuels for the temperatures investigated in the present study. Also, we would like to note that propargyl radicals +benzyl radical, and indenyl radical +CH₃ reactions are equilibrium driven and so have been described in a detailed reaction scheme in the current mechanism according to the findings of Matsugi and Miyoshi, and Zhao et al. [43,53]. Such a detailed description of these reactions is missing in the literature mechanisms, and rate used for both these reaction systems in literature mechanisms are significantly different from the recommendations of Matsugi and Miyoshi, and Zhao et al. [43,53], and has been noted by Matsugi and Miyoshi in their work. Finally, the current mechanism is informed by recent theoretical studies

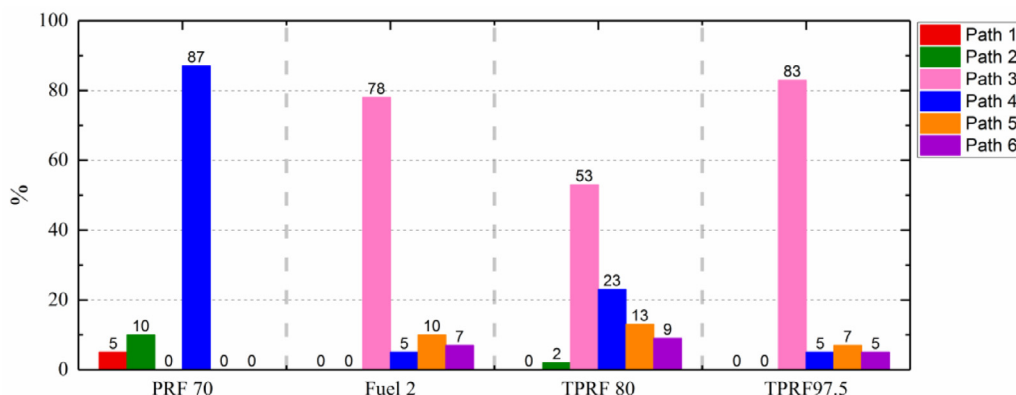


Fig. 16. Contribution of six pathways towards formation of benzene for the selected gasoline surrogates. Analysis conducted at 1150 K temperature for conditions listed in Table 1.

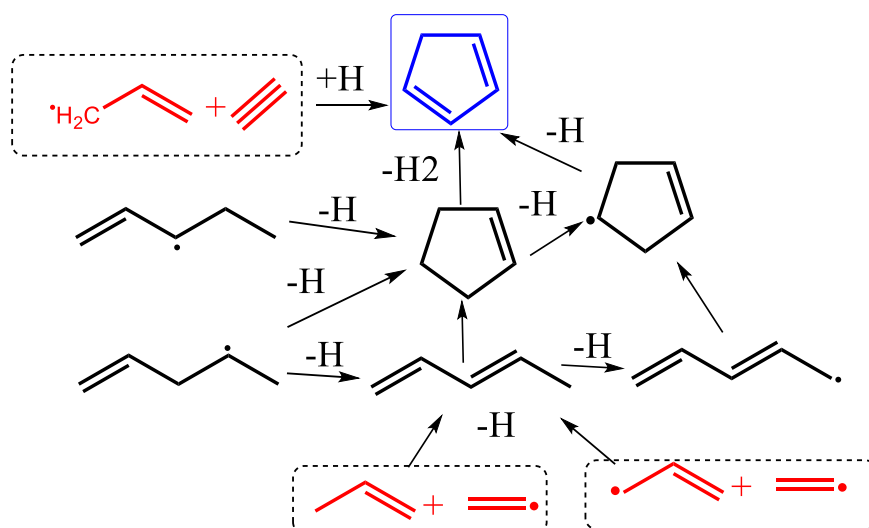


Fig. 17. Formation of cyclopentadiene from C_5H_9 radicals and C_2+C_3 reaction network.

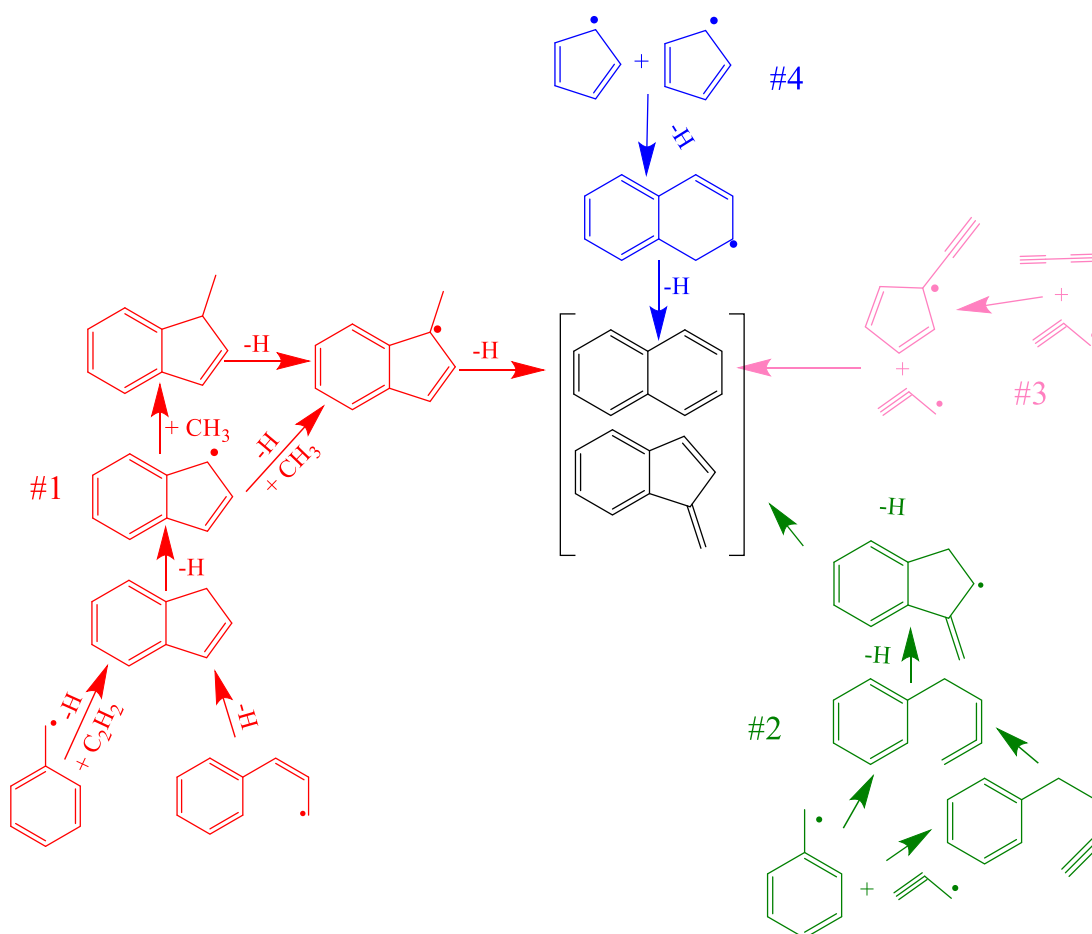


Fig. 18. Important reaction pathways for production of naphthalene at 1150 K.

(see Table S1 for more detail) that used state-of-the art techniques for molecular growth reactions including C_2 – C_5 species. Including the reactions of these small hydrocarbons with phenyl and benzyl radicals furthers the development of predictive fundamental mechanisms for PAH and soot. We acknowledge the mechanism still has shortcomings and further refinements are needed. However, more fundamental information from experiments and theory are

needed to better understand the dominant PAH growth pathways and their reaction rate constants. Some of the important reaction systems recommended for new, or more advanced, studies include: propargyl+ but-2-yne-1-yl radical; but-2-yne-1-yl radical+ but-2-yne-1-yl radical; $\text{C}_6\text{H}_5 + \text{C}_2\text{H}_4$; 1-naphthyl radical+ $\text{C}_3\text{H}_3/\text{C}_3\text{H}_4$; $\text{o-C}_6\text{H}_4\text{C}_2\text{H} + \text{C}_2\text{H}_4$; indenyl radical+ C_3H_3 ; α -methylnaphthalene radical+ C_2H_2 ; and $\text{o-benzyne} + \text{C}_2\text{H}_2/\text{C}_2\text{H}_4/\text{C}_3\text{H}_4/\text{C}_3\text{H}_3$.

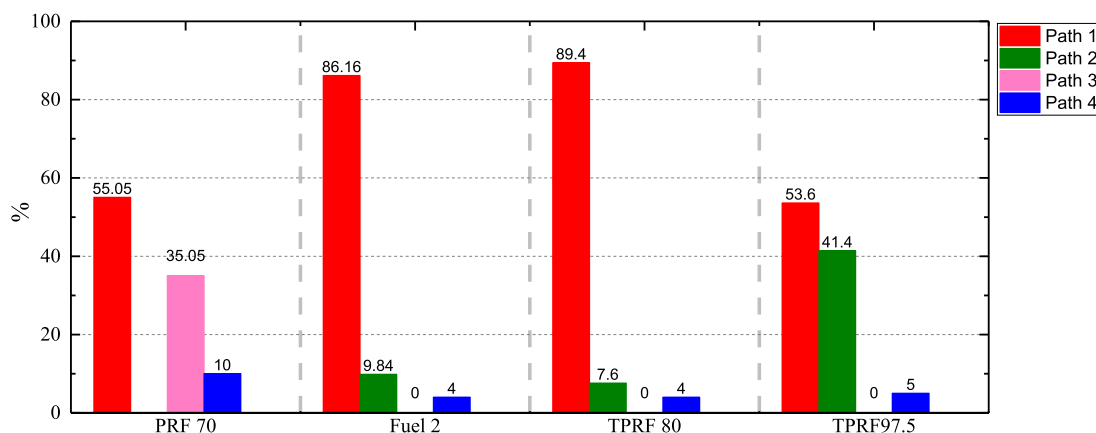


Fig. 19. Contribution of four pathways towards formation of naphthalene for the selected gasoline surrogates. Analysis conducted at 1150 K temperature for conditions listed in Table 1.

5. Conclusions

This work investigated the PAH formation characteristics from the pyrolysis of gasoline surrogates, such as binary and ternary mixtures of toluene, *n*-heptane and *iso*-octane. A detailed PAH kinetic model for gasoline surrogates was developed, and it successfully captured the concentration of major PAHs, as well as the small species produced in pyrolysis of gasoline surrogates. A synergistic effect was observed in *n*-heptane/toluene mixtures for both indene and naphthalene production, and captured by the kinetic model. Benzyl and acetylene were important for PAH formation in *n*-heptane/toluene mixtures. TPRFs produced more PAHs than PRFs because of the addition of toluene. In the PRF mixture, acetylene, propyne and cyclopentadienyl, produced by the unimolecular decomposition reaction of *n*-heptane and *iso*-octane, played an important role in the formation of PAH. In addition to acetylene and propyne, benzyl is another vital intermediate of PAH growth in TPRFs. This work provided new insights into the role of gasoline surrogate mixture composition on the PAH formation and the relevant reaction pathways. Such results may be used for optimizing fuel design and tailoring fuel/engine interactions for clean combustion technologies.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.combustflame.2020.06.001.

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